

CHINOOK DOCUMENTATION

ANALYSTLAB AFRICA

Week 3: SQL & Data Querying

Dataset: Chinook Database (Music Store)

1. DATABASE SETUP

The Chinook database was downloaded from GitHub and opened as aSQLite file. It contains tables for Artist, Album, Track, Genre, Customer and Invoice, showing how a digital music store keeps track of its sales.

2. CORE QUERIES

I used SELECT, WHERE, and ORDER BY to filter invoices over \$10 and customers from Brazil. I used GROUP BY and HAVING with SUM and AVG to find total revenue per country and the number of tracks in each genre.

3. ADVANCED QUERIES

- JOINS: I joined 4 tables (InvoiceLine, Track, Album, Artist) to find total revenue per artist. I also used a LEFT JOIN to find customers who never made a purchase.
- Subqueries: I found customers who spent more than the average customer. I also found tracks that were never bought.
- Window Functions: I used RANK() to rank customers by spending in each country, and ROW_NUMBER() to find each artist's top 3 selling tracks.

4. KEY FINDINGS

- USA brings in the most revenue, followed by Canada and France
- Iron Maiden is the best-selling artist
- Rock is the most popular genre
- Some tracks have never been sold at all

5. QUERY OPTIMISATION

I added indexes on the columns used most in joins (CustomerId, TrackId, GenreId, ArtistId). This made the queries run faster.

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